

EDES 301

MyoFX Proposal

Oct 2nd
Zeyi Lyu

MyoFX: EMG-Based Expression Control for Guitar Effects

I want to build a wearable biosignal controller that uses muscle activity to control guitar effects in real time.

The first prototype uses a MyoWare EMG sensor to measure forearm muscle activation. The EMG signal is read by a PocketBeagle, converted into a normalized expression value, and sent to a Daisy Pod, which performs the real-time guitar audio processing.

MyoWare was selected because it provides an integrated EMG front end and simple analog outputs, which reduces the hardware complexity of the first prototype.

(If MyoWare is considered too expensive, we can find other good options to replace it)

Background Information:

MyoWare Hardware applications

<https://github.com/AdvancerTechnologies/MyoWare-Arduino-Library> (official library)

<https://github.com/melissayu03/myoware-emg-detector> (hardware signal acquisition)

<https://github.com/Araadh3111/Bionix> (EMG to servo control)

Background Information: EMG to Music

<https://www.hackster.io/upsidedownlabs/muscle-melody-play-music-with-your-muscle-movements-emg-c0a0c0>

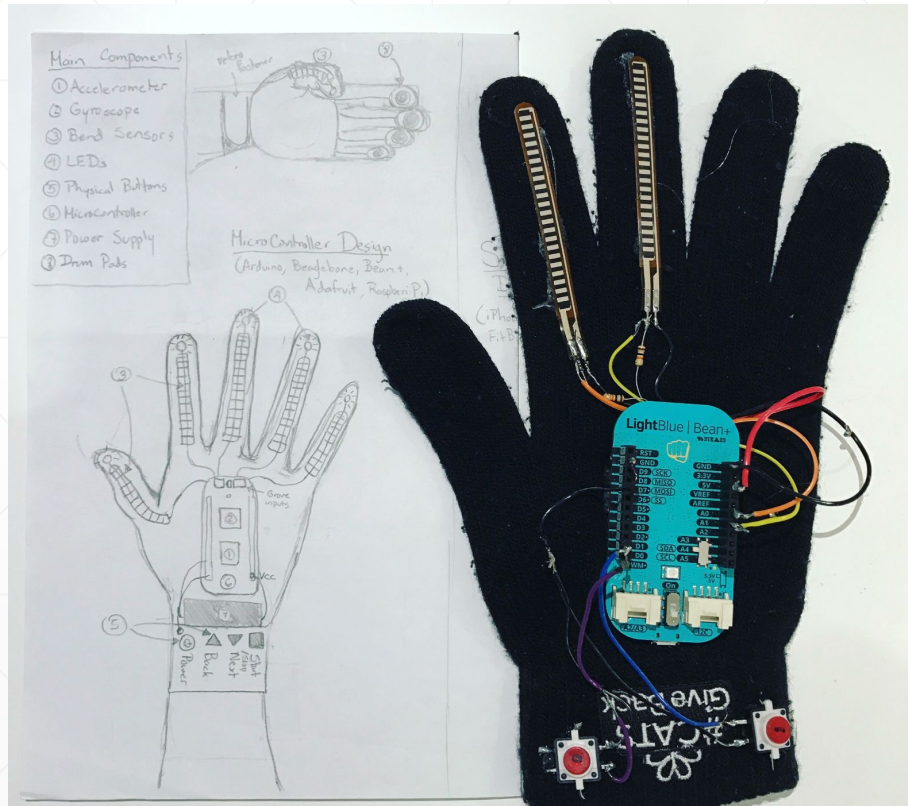
https://nime.org/proceedings/2025/nime2025_35.pdf



Background Information: Flex Bend

<https://github.com/interactiveaudiolab/DataGlove>

<https://github.com/sneak-thief/Sneaky-Gestures-V2>



Background Information: DaisySeed community

<https://community.daisy.audio/t/esp32-control-of-daisy-seed/1226> (how others build communications with other MCUs as DaisySeed's input)

<https://community.daisy.audio/t/daisy-serial-interface/1300/3> (serial problems)

<https://github.com/bkshepherd/DaisySeedProjects/tree/main/Software/GuitarPedal> (examples of MIDI CC input controlling DaisySeed)

Improvement: From Multi-Parameter Controllers to One Playable Tone Macro

Existing projects already show that the main parts are possible:

- MyoWare can measure muscle activity and turn it into a control signal.
- EMG music projects like EMMA show that muscle signals can be used for musical interaction.
- Daisy Seed projects show that external controls such as MIDI CC or knobs can change DSP parameters in real time.

But many of these systems use several sensors to control several different parameters.

For MyoFX, I want to use a simpler control architecture. I use one EMG signal to represent one overall expression level. That value can then change several effect parameters together instead of making the player control each one separately.

Improvement: From Multi-Parameter Controllers to One Playable Tone Macro

The main improvement is not adding more controls, but reducing the control complexity. Instead of asking the player to control several effect parameters separately, I want to use one EMG signal as one overall expression value and let that value change the guitar tone in a coordinated way.

The first goal is to test whether physiological input can become a predictable and playable musical interface, similar to how an expression pedal gives continuous control.

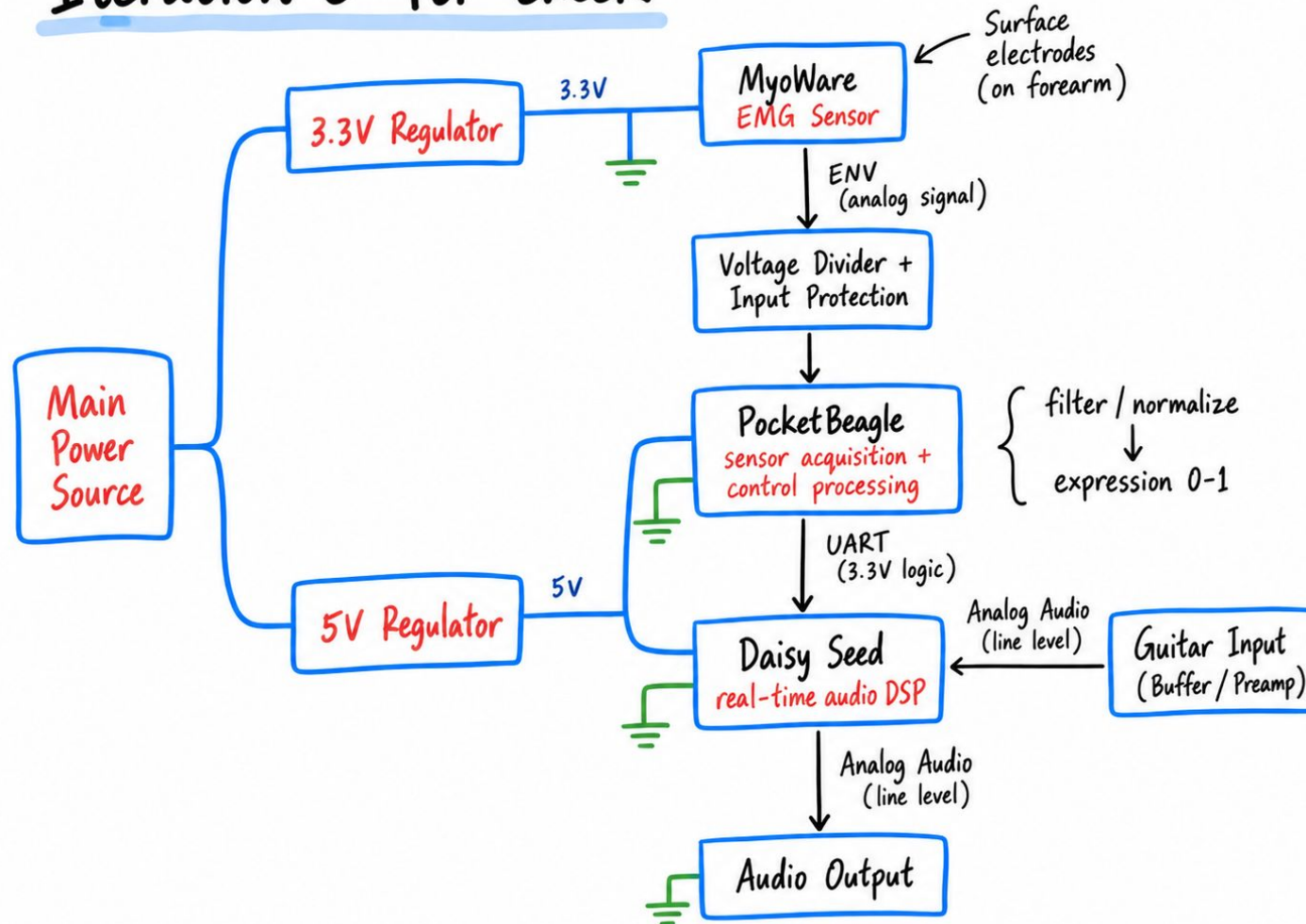
If this works, the next version can improve the system in two ways:

- add Bluetooth so the sensing side can become wireless and more wearable
 - add finger flex as a second control dimension for richer expression
-

Prototype to check feasibility

MyoWare also supports 5V supply, so I don't know whether using 3.3V would be better

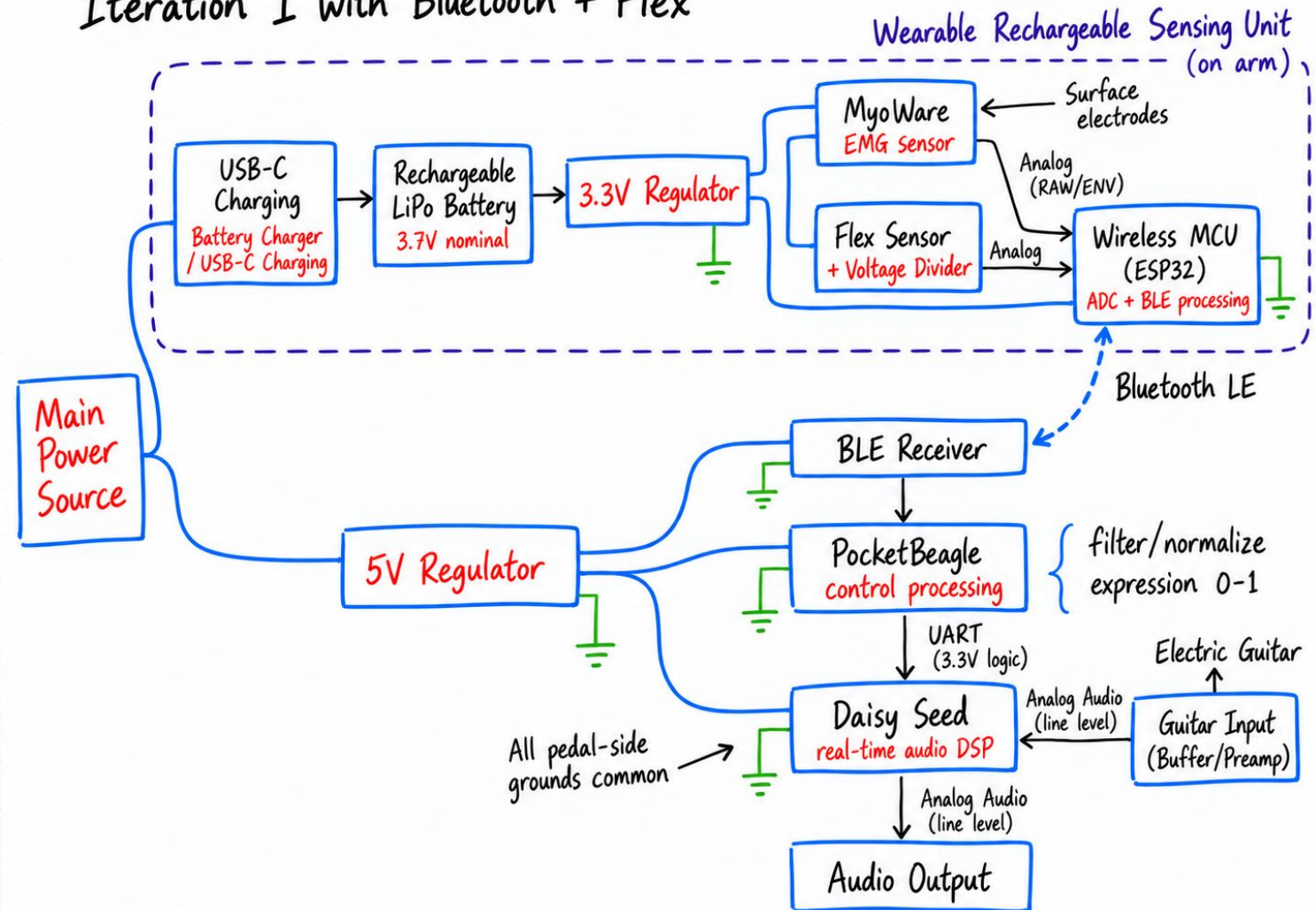
Iteration 0 for check



If the first prototype works:

I also want to explore what the final wireless design might look like. If the sensing side becomes wearable, I may need to use a smaller wireless MCU such as an ESP32 and eventually move away from the PocketBeagle. For this course, though, I want to keep Iteration 0 simple and use the PocketBeagle first, so I can focus on testing whether the EMG control itself actually works well.

Iteration 1 with Bluetooth + Flex



Components / Budget

Component?	EDES301 to Buy?	Cost
MyoWare 2.0 https://www.sparkfun.com/myoware-2-muscle-sensor.html	?	\$42.95
Daisy Pod	no	\$79
AgCl electrode https://www.sparkfun.com/disposable-surface-emg-ecg-ekg-electrode-24mm-10-pack.html	?	\$9.95/#10
USB 5 V cables/power	no	
(if PocketBeagle 3.3V not appropriate) 3.3V LDO	?	\$1.50
(if version2)BlueTooth device	?	?
(if version2)Flex Sensor 2.2" https://www.sparkfun.com/flex-sensor-2-2.html	?	\$12.50
(if version2)ESP32 Feather https://www.adafruit.com/product/5900 LiPo Battery	?	\$20